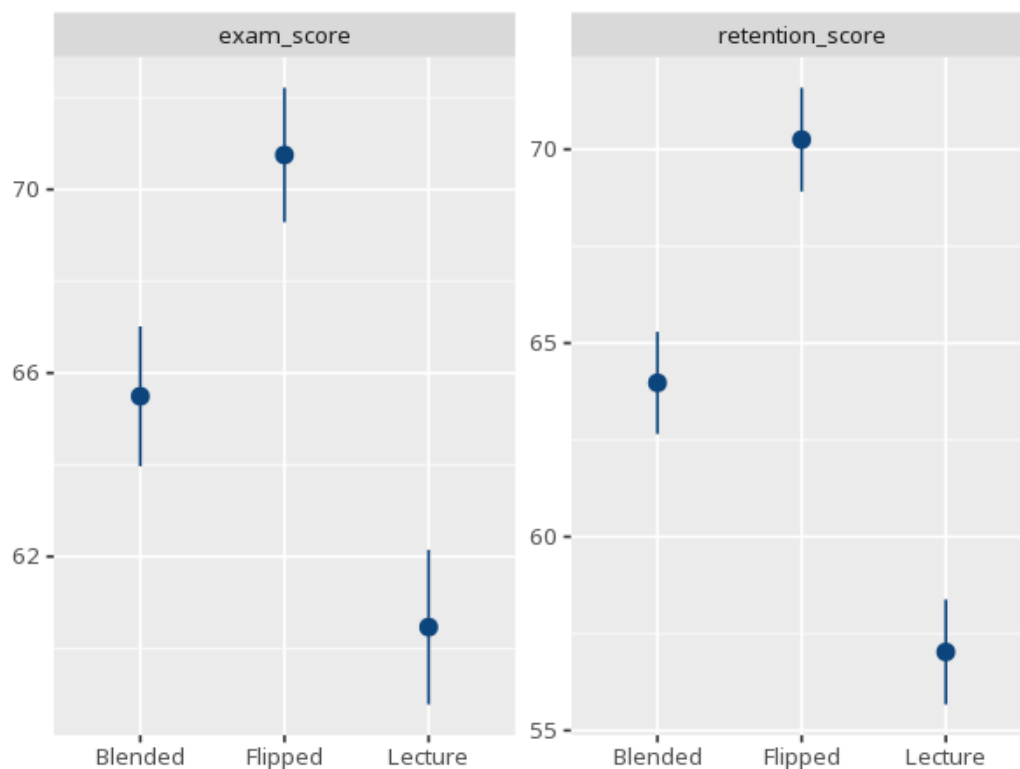


1 MANOVA

Effect	Pillai / Wilks	approx F	df1	df2	p	partial η^2 [95% CI]
teaching_method	0.46	35.03	4	474	<.001	0.23 [0.17, 1.00]

DV	F	df1	df2	p	partial η^2 [95% CI]
exam_score	43.07	2	237	<.001	0.27 [0.19, 1.00]
retention_score	97.01	2	237	<.001	0.45 [0.38, 1.00]



Tests whether groups differ on a set of outcomes jointly. Read the multivariate p (Pillai's trace is robust) first; if significant, the per-DV follow-up ANOVAs show which individual outcomes drive it. Because you run one ANOVA per outcome, correct those follow-up p-values for multiple comparisons (e.g. Bonferroni: divide alpha by the number of outcomes) before calling each significant. Your significance threshold (α) is 0.05.

A MANOVA gave Pillai's $V=.46$, $F(4,474)=35.03$, $p < .001$, partial $\eta^2=.23$ [0.17, 1.00].

Statistical basis: Pillai's trace multivariate test (headline statistic) (Pillai, K. C. S. (1955). "Some new test criteria in multivariate analysis." *Annals of Mathematical Statistics*, 26(1), 117-121.); Wilks' Lambda multivariate test (Wilks, S. S. (1932). "Certain generalized distributions in multivariate analysis."

sis." *Biometrika*, 24(3/4), 471-494.); Box's M test of homogeneity of covariance matrices (Box, G. E. P. (1949). "A general distribution theory for a class of likelihood criteria." *Biometrika*, 36(3-4), 317-346.); Multivariate effect size (partial η^2 from the multivariate approx. F) (Ben-Shachar MS, Lüdtke D, Makowski D (2020). "effectsize: Estimation of Effect Size Indices and Standardized Parameters." *Journal of Open Source Software*, 5(56), 2815.).